

MAS · Vector Calculus and Motion — Patterns and Solution Approaches

Source: 12 past exam questions from 2016–2025, marking keys + exam reports. Appears only in the Calculator-assumed paper. Original questions: see [topics/04_Vector_Calculus.pdf](#). Review status: has undergone one round of independent review and was revised according to feedback (see [Review_log.md](#)). On 2026-10-09, following user feedback, 2017A-Q18 and 2019A-Q18, which do not involve vectors, were moved out of this topic (into "Rates of Change and Differential Equations").

Marking panel reminders

- **Difference between distance travelled and displacement vector** (2025 report focus): $\int_a^b \mathbf{v}(t) \, dt$ is the **change in displacement vector** (has direction), while $\int_a^b |\mathbf{v}(t)| \, dt$ is the **distance travelled** (scalar).
- **The constant of integration is a vector** (2018 report: many people cannot clearly show how to use the constant of integration to obtain $\mathbf{r}(t)$): $\mathbf{r}(t) = \int \mathbf{v} \, dt + \mathbf{c}$, then use $\mathbf{r}(0)$ to solve for each component of $\mathbf{c}$.
- **Notation**: Vectors must be labelled (with a wavy underline or bold), and velocity vector and speed must be written separately; the keys for 2019A-Q13 and 2022A-Q10 each have 1 mark specifically for notation.
- After eliminating the parameter to obtain the Cartesian equation, **you must state the value restrictions** (often both x and y need restrictions) (the keys for 2016A-Q16, 2024A-Q18 and 2025A-Q12 all have this mark).

Formula sheet

Quantity	Expression
Velocity	$\mathbf{v}(t) = \mathbf{r}'(t)$ (differentiate component by component)
Acceleration	$\mathbf{a}(t) = \mathbf{v}'(t)$
Speed	$ \mathbf{v}(t) = \sqrt{\dot{x}^2 + \dot{y}^2 (+ \dot{z}^2)}$
Distance travelled (arc length)	$\int_a^b \mathbf{v}(t) \, dt$
Change in displacement	$\int_a^b \mathbf{v}(t) \, dt = \mathbf{r}(b) - \mathbf{r}(a)$
Direction of motion	angle with the x -axis $\tan^{-1}(\dot{y}/\dot{x})$, note the quadrant

Pattern 1: Given $\mathbf{r}(t)$, find velocity, acceleration and speed

Approach: Differentiate component by component $\rightarrow$ substitute the time $\rightarrow$ take the magnitude for speed $\rightarrow$ draw the vector on the diagram (tail at the position at that time, arrow pointing in the direction of motion).

Constant speed check (2020A-Q12): $\mathbf{r} = (10\cos 0.1t, 10\sin 0.1t, 0.2t) \Rightarrow \mathbf{v} = (-\sin 0.1t, \cos 0.1t, 0.2) \Rightarrow |\mathbf{v}| = \sqrt{\sin^2 + \cos^2 + 0.04} = \sqrt{1.04} \approx 1.02 \text{ m/s}$, independent of $t \Rightarrow$ **the speed is constant** (constant speed), but the direction of the velocity vector is always changing, so it is not uniform straight-line motion. You must state that $\sin^2 + \cos^2 = 1$ is used (1 mark separately). Observation deck height = $0.2 \times 65 = 13 \text{ m}$.

Past exam questions: 2016A-Q16(b), 2017A-Q15(a)(d), 2019A-Q13(a)(b), 2020A-Q12, 2023A-Q16(a), 2024A-Q18(b)(d).

Pattern 2: Given $\mathbf{a}(t)$ or $\mathbf{v}(t)$, integrate to find $\mathbf{r}(t)$

Standard procedure

1. Integrate component by component, adding the **constant vector** $\mathbf{c} = (c_1, c_2)$.
2. Use the initial conditions ($\mathbf{r}(0)$, $\mathbf{v}(0)$) to solve for each component.
3. Write the full $\mathbf{r}(t)$.

Projectile motion, two models:

- Gravity + wind (2018A-Q19): $a = (0, -9.8) \Rightarrow v = (70\cos\theta + w, 70\sin\theta - 9.8t) \Rightarrow r = ((70\cos\theta + w)t, 70\sin\theta \cdot t - 4.9t^2)$. Wind affects only the horizontal component.
- Horizontal drag + vertical lift (2021A-Q19): launched horizontally from $E(100, 120)$; $x' = 32e^{-0.05t}$; vertical acceleration $h'' = s - 9.8$ (s is the lift), use the given r to solve backwards for $s = 4.8$, so $r = (740 - 640e^{-0.05t}, 120 - 2.5t^2)$. **Range or landing point:** $y(t) = 0$ (or equal to the slope height) $\Rightarrow$ solve for $t \Rightarrow$ substitute into $x(t)$. Maximum range $\Rightarrow$ differentiate with respect to θ or use CAS to find the maximum. **Landing angle:** use the angle between the velocity vector at the instant of landing and the slope or horizontal plane (2021A-Q19(e)).

Past exam questions: 2017A-Q15(c), 2018A-Q19(a)(c), 2021A-Q19, 2022A-Q10(b).

Pattern 3: Eliminate the parameter to obtain the Cartesian equation ★

Common identities

- $\cos^2 t + \sin^2 t = 1 \Rightarrow$ circle or ellipse (2022A-Q10(c));
- $\cos 2t = 1 - 2\sin^2 t$ or $2\cos^2 t - 1$, which directly gives the relationship between x and $y \Rightarrow$ a segment of a parabola (2016A-Q16: $x = 2y^2 - 4$);
- First use double-angle formulas to turn the two components into the same angle, then use $\sin^2 + \cos^2 = 1$ to eliminate the parameter $\Rightarrow$ a general curve (2019A-Q13(d): $(1 - y)^2 + (x^2/2 - 1)^2 = 1$, a figure-eight shape);
- x is linear in t : solve directly for t from x and substitute into y (2018A-Q19(b); the 2018 report specifically identifies this part as poorly answered);
- $(e^t + e^{-t})^2 - (e^t - e^{-t})^2 = 4 \Rightarrow$ hyperbola $x^2 - y^2 = 4$ (2024A-Q18(e));
- Exponential parameter: solve x for $t = \ln(\dots)$, then substitute into y (2025A-Q12(a)).

The final step must not be missed: state the range of the actual path, **derived from the range of t** , key wording: 2016A-Q16: $-2 \leq y \leq 2$ (or $-4 \leq x \leq 4$); 2024A-Q18: $x \geq 2$ **and** $y \geq 0$ (writing only $x \geq 2$ includes the lower branch and loses marks); 2025A-Q12: $1 \leq x \leq e^3$.

Past exam questions: 2016A-Q16(a), 2018A-Q19(b), 2019A-Q13(d), 2022A-Q10(c), 2024A-Q18(e), 2025A-Q12(a).

Pattern 4: Distance travelled (arc length) and interpretation of displacement

Approach

1. Find the start and end times. **The time for one complete circuit = the least common multiple of the periods of the components:** 2022A-Q10 is 2π , 2019A-Q13 is 4π (contains $t/2$), 2017A-Q15 is 6π (contains $t/3$). These two questions each have 1 mark for the correct limits. You can also work backwards from the coordinates to find t : point B is $(10/3, 8/3)$ when $t = \ln 3$.
2. Write $\int_a^b \sqrt{(\dot{x})^2 + (\dot{y})^2} dt$ (the usual mark-earning points: start/end t -values or limits, speed expression, evaluated result or notation; allocation varies by year; "Do not evaluate" questions have no mark for evaluating).
3. When the question says "Do not evaluate", just write the integral expression containing trigonometric functions.

Explain the meaning of the integral (2 marks)

- $\int_0^1 v(t) dt$: "the change in displacement (position vector) during the first second".
- $\int_0^{2\pi} |v(t)| dt$: "the distance travelled in one complete circuit (2π seconds)" (2022A-Q10).
- $\int_0^3 r'(t) dt$: the change in displacement vector, drawn on the diagram as an arrow from $r(0)$ to $r(3)$ (2025A-Q12(c)).

Past exam questions: 2016A-Q16(c), 2017A-Q15(b)(e), 2019A-Q13(c), 2022A-Q10(a), 2023A-Q16(b), 2024A-Q18(c), 2025A-Q12(b)–(d).

Pattern 5: Closest distance between objects moving in straight lines (3D)

Approach

1. Write the positions of the two objects: $r_K(t) = r_{K0} + t \cdot v_K$, $r_B(t) = r_{B0} + t \cdot v_B$.
2. Relative position $d(t) = r_K(t) - r_B(t)$. The position of P relative to Q = $r_P - r_Q$; when the question asks for a relative position vector, the order must be correct (2017A-Q14(b): "from the top of the tower" $\Rightarrow r_{\text{drone}} - r_{\text{tower}}$, with 1 mark in the key). When only finding distance, the order does not affect the magnitude.
3. Closest distance: set $d(t) \cdot (v_K - v_B) = 0$, or use CAS to find the minimum of $|d(t)|$.
4. "Time period within 50 m": solve $|d(t)| < 50$ and find the length of the time interval (2017A-Q14(c)).
5. Shortest distance between two paths (2024A-Q16(c): **velocity and departure time** can be adjusted, so the problem reduces to the shortest distance between two straight lines): use two independent parameters λ , μ and require the common perpendicular vector to be perpendicular to both direction vectors (2024A-Q16(c)).
6. Angle of descent (angle with the horizontal plane): first find the angle ϕ between the velocity vector and k , then angle of descent = $|90^\circ - \phi|$ (for Ben's $v = (-0.5, 1, -1.5)$, $\phi \approx 143.3^\circ$, $90^\circ - \phi = -53.3^\circ$, answer 53° ; 2024A-Q16(b)). You can also use $\sin^{-1}(|v_z| / |v|)$ directly.

(Ferris wheel and carousel questions (2017A-Q18, 2019A-Q18) do not involve vectors and have been moved into the related rates pattern in the "Rates of Change and Differential Equations" notes.)

Past exam questions: 2017A-Q14, 2024A-Q16.
